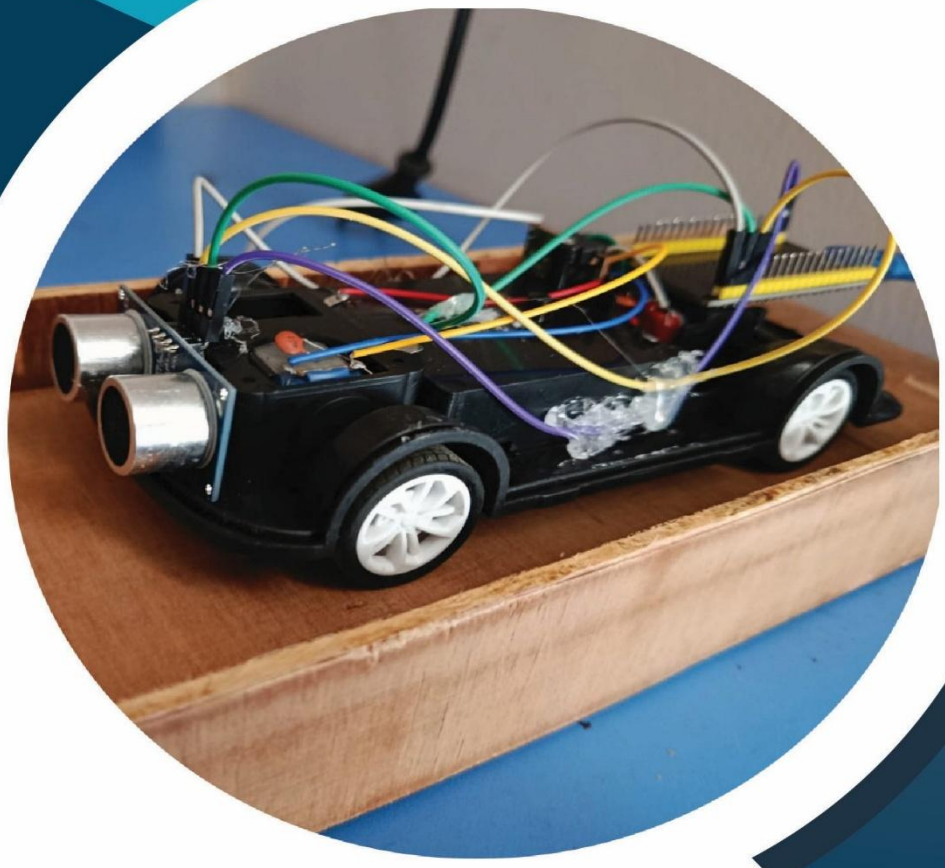


30207

Technometer

New Cairo STEM School

2026/2025

semester 1

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Chapter I

(Present and Justify a Problem and Solution Requirements)

I. Egypt Grand Challenge

II. Problem to be solved

III. Research

IV. Prior solution

I. Egypt Grand Challenges:

With the constant growth of our motherland, it faces progressively harder obstacles and more challenging problems. These problems are usually referred to as “The 11 Egyptian Grand Challenges.

Nowadays, Egypt stands at a critical crossroads, facing a set of interconnected problems and challenges that will shape its future. The continued population growth, without proper land and resource usage, alongside the constant need for urbanization and staggering economic ambitions, poses a significant threat.

All of this takes place while Egypt struggles to cope with new emerging problems and challenges, such as global warming, which has recently affected the world, or the need to adopt new and improved processes and methods for producing energy instead of the traditional carbon-emitting means and methods used to generate electricity and energy. These Grand challenges represent the key obstacles that are facing Egypt soon and the long-term run and sustainability of the country.

Each challenge requires innovative solutions, strategic planning, and collective efforts from everyone in the country, including the government, which should support the project's efforts, as well as politicians, business owners, scientists, and citizens alike, even students like us. Each person's effort is important for our beloved country, paving the way for a stronger, greener, and more resilient future.

The first step to solving the problem is to know what the problem is, which is why we have mentioned the 11 Grand Challenges that Egypt faces, and to solve them, making Egypt more developed:

- Improve the use of alternative energies.
- Recycle garbage and waste for economic and environmental purposes.
- Deal with urban congestion and its consequences.
- Work to eradicate public health issues/diseases.
- Increase the industrial and agricultural bases of Egypt.
- Address and reduce pollution fouling our air, water, and soil.
- Improve the use of arid areas.

- Manage and increase the sources of clean water.

- Deal with population growth and its consequences.

- Improve the scientific and technological environment for all.

- Reduce and adapt to the effects of climate change

❖ Improve the scientific and technological environment for all:

Science and technology are one of the basic foundations of any country, especially for developing countries, to be on the right path of growth, so Egypt has tried to be on the track as a developing country that aims to improve scientific levels and apply the latest technological applications. For having the suitable ecosystem to support the enhancement of our scientific degree, we should consider several factors that are included in the scientific process: for instance, the scientific research level, the educational system, scientific infrastructure, and funding. Egypt faces many obstacles in most of these scientific factors, so its scientific and technological environment isn't the most suitable one for the researchers to get a good outcome during the scientific process, but many efforts are being made to improve the scientific and technological environment, such as:

Egypt Vision 2030:

It is a vision launched by the Egyptian government. It aims for development in all the country's sectors, but it gives priority to science, innovation, and digital transformations as the key to economic growth.



Figure 1

Funding:

In recent years, the government has increased financial support for research institutions and scholarships to aid brilliant minds.

Educational campaigns:

-Zewail **City of Science and Technology,** which is considered a hub of innovations and advanced scientific research.

-**Smart Village** near Cairo hosts technology companies, startups, and universities working together on AI, ICT, and renewable energy projects.

-**Egyptian Knowledge Bank (EKB)** provides free access to millions of international journals and scientific databases for all citizens.

Educational programs:

Egypt's first STEM school was established in 2011 as a pioneering project by the Egyptian Ministry of Education, with USAID support, to foster innovation in science, technology, engineering, and math education. Many STEM schools opened soon after to encourage innovation, problem-solving, and creativity among high school students.



Figure 2

Improving these factors will indeed facilitate the scientific process, and it is expected that Egypt will have reliable research institutes and highly ranked researchers across all scientific fields and technological disciplines.

❖ Address and reduce pollution fouling our air, water, and soil:

The term “pollution” means: the presence of a substance in or into the environment which has harmful or hazardous effects, according to the Oxford Languages dictionary.

Pollution could be seen as a key player in shaping the sad reality of our beloved country, as more pollution means fewer available resources, which could make Egypt vulnerable to some serious problems, such as:

- **Regional disasters:** Famines could initiate as less farmable lands mean less crops, leading to food shortages in developing countries such as Egypt.

- **Health hazards:** As…

- Air pollution is a primary cause of immunity deficiency and increases the chance of catching bodily issues like pulmonary cancers and other pulmonary diseases like influenza, and allergies due to the inhalation of polluted air containing CO_2, CO, and other harmful gases.

- Anybody could consume contaminated water that contains a harmful substance like lead or arsenic, and maybe even a harmful micro-organism like malaria plasmodium or bilharzia.

- **Economic problems**: it also harms the economy as it could affect our valuable resources like water, air, and the fertile soil used for producing essential crops, leading to a subtle increase in the prices of our basic products, and this could lead to potential wars about something we take for granted, like wheat.

- **Global warming & Climate change:** As **we** all know, greenhouse gases like CO_2, CH_4, and other gases presented in Figure 3, which are caused primarily by air pollution, form a layer around the earth that allows white light to pass through it, but traps infrared rays that cause the gradual rise in the average temperature of the earth.

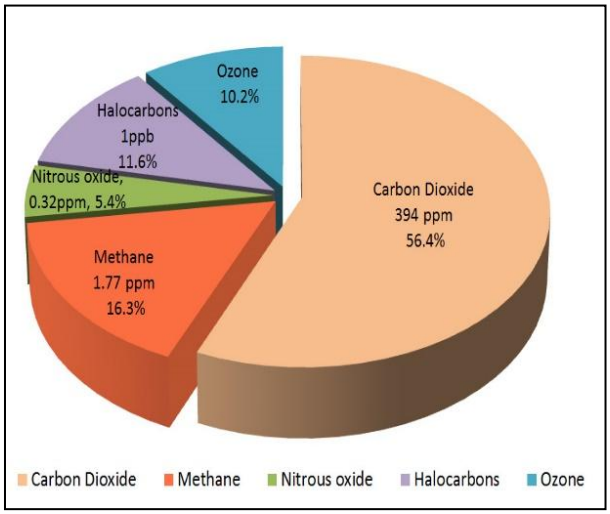


Figure 3

This increase causes various problems, like:

- **Melting ice caps**:
  The polar ice caps melt as the temperature increases, causing the rise of the ocean level, leading to some low-lying regions like Alexandrea to sink underwater.

- **Climate change:**
  As the average temperature of the globe is constantly rising, the weather becomes more extreme as winters become colder and summers become hotter. This affects the environment and the ecosystem.

Some arctic animals, like penguins, cannot survive in the hot biomes or in hot weather, and African Elefants cannot endure the harsh winters. This shift could result in the extinction of some animals or even whole species.
Some crops need special requirements to grow, like rice, needing excessive amounts of water and sunlight to grow. If the climate shifts slightly, this could make the biome that used to grow these crops incapable of producing them, leading to the problems mentioned before.

In summary, pollution causes various problems like health hazards and famines, poses a potential economic threat, and endangers the global ecosystem. A problem with a horrendous impact like this needs to be stopped, no matter the cost.

❖ Recycle garbage and waste for economic and environmental purposes:

The environmental necessity of waste management and recycling regarding its economic effect. Waste generation is an unavoidable product of human civilization, connected deeply to the social and economic aspects of civilians' lives.
As global economies transition from linear to circular models, the effective management, recovery, and recycling of waste streams have become critical for both environmental and economic sustainability. Effective action is necessary to achieve ambitious goals, such as the EU's 55% target for plastic packaging waste recycling by 2030, which addresses high dependence on fossil feedstock, low

recycling rates, and significant leakage of plastics into the environment. Developing innovative technologies like chemical and combined physical recycling is key to securing material resources, increasing recycling rates, and potentially improving environmental performance, especially by handling waste flows otherwise sent to disposal or incineration.

The Environmental Performance and Life Cycle Assessment of Waste Management:

Environmental assessment tools, specifically Life Cycle Assessment (LCA), are employed to evaluate the performance of Municipal Solid Waste (MSW) management systems and to quantify the potential environmental impacts resulting from different waste treatment methods. LCA models investigate waste management options by computing impact levels based on detailed unit processes, quantified values of various resources, and emissions.

For waste management systems, the choice of preferred management option is based on three main criteria. The first of these is the maximization of material recovery while minimizing processing impacts, primarily related to energy consumption. This criterion directly reflects environmental performance and is in line with the waste hierarchy.

Comparison of Recycling and Energy Recovery Studies comparing recycling technologies (mechanical, physical, chemical) with energy recovery (incineration) for specific waste streams, such as plastic waste, reveals several critical findings:

- **Climate Change (Global Warming Potential - GWP):**
  Recycling is generally preferable to energy recovery in all pathways analysed for climate change, notably for mixed polyolefin waste that is not currently mechanically recycled. This is because the environmental savings from energy recovery are often insufficient to compensate for the environmental impacts of waste incineration and its associated emissions. For example, the management of one tonne of mixed shredded plastics from large WEEE (cooling and freezing

appliances) resulted in net savings for mechanical recycling (-1522 kg-eq./t) but net burdens for energy recovery via incineration (1095 kg -eq./t).

- **Other Environmental Categories (Acidification and Particulate Matter):** While maximizing material recovery holds true for categories like ozone depletion and fossil resource use, for some categories, including acidification, particulate matter, and eutrophication, energy recovery can perform better than energy-intensive recycling pathways. This counter-performance is largely because current incinerators in Europe, on average, achieve much lower emissions of substances like dust than the average energy production mix used for calculating the impacts of recycling's energy consumption. The shift to a cleaner European energy mix is expected to further increase the gap in favour of recycling, supporting the validity of the EU waste hierarchy.

- **Municipal Solid Waste (MSW) Management:**

  In the context of MSW, which includes collection, transportation, and disposal, a "Conjunctive Disposal" scenario (composting and landfilling) was found to yield minimum environmental impacts compared to a "Business as Usual" scenario (collection, transportation, landfilling) and a "Recycling" scenario, based on impact metrics including GWP, Acidification Potential (AP), and Eutrophication Potential (EP) Economic Viability and Technological Imperatives.

  The decision on the preferred waste management option must also consider the technical feasibility (specificity of the plastic waste stream and treatment required) and the economic feasibility.

- **Economic Factors in Plastic Recycling**

  The economic viability of physical and chemical recycling, particularly without public support, is determined by three main parameters:

  feedstock prices, capital expenditures (CAPEX) and operational expenditures (OPEX), and output prices.

- **Cost and Revenue:**

Preliminary economic data suggests some physical and chemical recycling technologies, such as dissolution and glycolysis, may have already reached break-even status in terms of costs, although other technologies like metanalysis, pyrolysis, and gasification were found to have negative net incomes. However, this current analysis does not account for possible gate fees or other cost correction measures.

- **Future Projections:**
  Physical and chemical recycling technologies are still experiencing rapid development, and their costs are projected to decrease in the future. Simultaneously, costs for mechanical recycling are expected to remain stable, while costs for virgin plastics production are expected to substantially increase with increasing fossil fuel prices. This shifting in the economic landscape is estimated to lead to positive earnings for major chemical recycling technologies, like metanalysis by 2025, pyrolysis by 2033, and gasification by 2040.

- **Cost Efficiency and Alternative Energy Recovery:**
  The environmental ranking of waste management scenarios can often correlate with economic performance. For MSW management in Kathmandu, the Conjunctive Disposal scenario (composting and landfilling) that yielded minimum environmental impacts was also identified as a cost-efficient option. Furthermore, waste management offers a pathway to sustainable energy recovery. The anaerobic digestion of organic waste combined with livestock manure, which is part of effective waste management, is highlighted as a sustainable and environmentally friendly energy option for producing biogas. Fuel Energy Consumption (FEC) itself is a key metric, calculated based on the monetary value expended to purchase fossil fuels (petrol, diesel, and kerosene) used in the collection, transport, and management of waste.

In conclusion, the effect of waste recycling serves the double purpose of maximizing material recovery—aligning with the waste hierarchy and achieving environmental savings in climate change—while simultaneously creating economically viable industrial sectors projected to reach profitability in the medium to long term, thereby fostering a robust circular economy.

❖ Reduce and adapt to the effect of climatic change:

In Egypt, climate change is starting to feel like more than just a political tool. It's not just the weather; it is impacting people's lives, their jobs, what they eat, and where they live. and the average temperature in Egypt has risen by approximately 1.5°C during the past century. Estimates suggest it may climb by 2–3°C more by 2050, resulting in increasingly frequent and severe heatwaves. As shown in Figure 4:

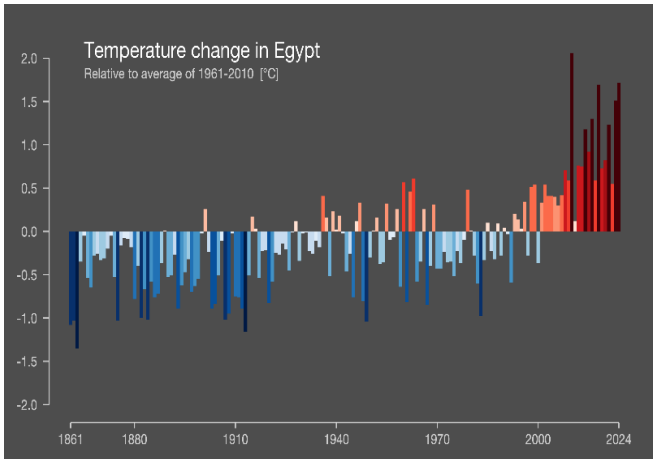


Figure 4

What contributes to climate change?

There are numerous reasons for this challenge, such as:

- **Burning Fossil Fuels:**

  When we use coal, oil, or gas to power our homes and factories, or to drive our cars or cook food on a stovetop, what we do is release gases like carbon dioxide into the air. These gases trap heat around the Earth, making our planet warmer over time.

- **Cutting Down Trees (Deforestation):**

  Trees are the lungs of our planet. They inhale carbon dioxide and exhale clean air. But when forests are cut down or burned down, the carbon that has been stored by trees is released back into the atmosphere, and we lose one of our best natural defences against climate change.

- **Industrial Pollution:**

Factories that produce the things we use every day, from clothing to cement, also generate gases, including carbon dioxide, methane, and nitrous oxide. These are potent greenhouse gases that exacerbate global warming.

II. Problem to be solved:

Egypt faces a growing recycling problem that affects both people and the environment. Every day tons of garbage are produced and small amounts of it only recycled it will affect the environment and produce pollution in the air and in the Nile and this will affect people before the country so if we used this garbage from old electronics like things that not recycled easily and benefit from it to put it in a device to produce another device from the old components will benefits the country technology and decrease the wastes and help the country. This is not only about cleaning, but also about people's health and pure air, and it will make a brighter future.

⇒ If the problem was solved:

- **The Nile River will be clean again:** The water quality of 24 sites between Aswan and Cairo along the Nile was investigated. To evaluate the suitability of water for aquatic life and drinking purposes, the indices of water quality (WQI), heavy metal pollution (HPI), and contamination (C_d) were computed. The water quality variations were primarily related to inorganic nutrients and heavy metals, with sites affected by an intensive load of urban, agricultural, and industrial wastewater exhibiting a serious deterioration in water quality compared to other sites.

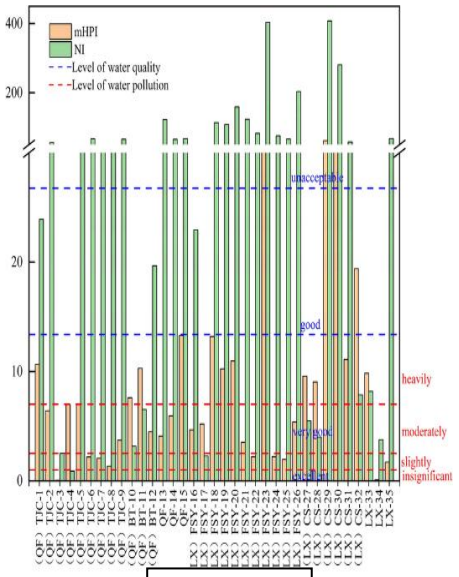


Figure 5

These heavy metals are directly associated with electronic waste; for example, the excessive amounts of copper, iron, tin, and silicon in the Nile River, as shown in Figure 5.

- **The technological environment will flourish:**

  As the number of technological devices and scientific knowledge. As increases occur, technological improvements will soon follow. The "waste" could potentially be beneficial. As the broken electronic devices are used in manufacturing other devices using their working components, the amount of silicon, copper, tin, and other important materials will enter the cycle again.

⇒ **If the problem was not solved:**

- The Nile situation will worsen:
  According to the data shown in Figure 5, the "Heavy metal" percentage in the Nile water increased over the years, and if this problem is not solved by either recycling or reusing these materials, it will continue to deteriorate.

- The "E-waste" will not be processed:
  As these E-devices need processing to re-enter the manufacturing cycle, having insufficient numbers of workers to extract the important components (e.g., microcontrollers) will lead to wasting a lot of useful hardware.

- The waste will remain waste:
  The useful metals that are mentioned above are abundant in the E waste, and not having an effective way to reuse them will lead to them becoming potential pollutants.

III. Research:

Thorough research was conducted on the topics related to the project, which were:

- The advantages of recycling
- How we will recycle electric devices
- What are the advantages of recycling electric devices
- How we will use old parts to make a new device
- If we make a new device from old parts, it will reduce electric waste.
- People know about the dangers of electric waste

- **How we will use it in the capstone project:**
- Firstly, about recycling garbage:

  Recycling turns waste materials into new, usable products, reducing environmental impact.

  Electronic waste contains valuable parts like circuit boards, motors, and sensors, which can be reused to build new devices.

- Secondly, about the electric waste:

  By repurposing these components, we can create functional gadgets while saving resources, lowering costs, and promoting sustainable technology.

  And about the electric waste, it is very dangerous to the environment and people so we can use it instated to throw it in the seas or in oceans or the factory we are melting it and emitting gas to pollute the environment

  So, we can use the electric waste to make a new device like you use old parts and know the efficiency of the parts and use it to make a new device also we must tell the people about that not only our project so we must make announcements to help people to reduce garbage and improve recycling so people can reduce electric waste and make new devices with those old parts.

1) Topics related to the solutions:

1. **Electrical and electronic devices**:

What are electrical and electronic devices?

- Electrical devices:
  - An electrical device comprising conductive components, such as steel, aluminium, iron, etc., designed to transform electrical energy into an alternative form of energy. Take a motor, for instance; when energized, it efficiently converts electrical energy into mechanical energy. Subsequently, the motor can be utilized to propel various objects**.**
  - Electrical devices operate on Alternating Current (AC). The AC power generated can be single-phase or three-phase, starting from 220V to 440V within the 'V' to 'kV' variety.
  - Devices within the electric area basically produce voltage and are modern. These devices play a critical role in strength systems, offering vital energy for diverse applications.
  - Traditional electrical devices cannot make choices independently. For example, an electrical bulb truly responds to the presence or absence of electrical strength.

❖ Components of electrical devices:

Electrical devices have common components that participate in performing specific functions of this device; all of these components are involved in the electric circuit of this device. These components are:

1. **Resistor:**

A resistor is one of the most fundamental electrical components. It resists the flow of electric current, limiting or controlling the amount of current that passes through a circuit.

- **Function:**
  Resistors are primarily used to control voltage and current levels within a circuit. They dissipate energy in the form of heat and help protect sensitive components from receiving too much current, which could cause damage.



Figure 6

2. Capacitor:

A capacitor is a device that stores electrical energy in an electric field. It temporarily holds electric charge, then releases it when needed.

- **Function:**
  Capacitors are used for storing energy, filtering signals, and stabilizing voltage and power flow. They smooth out fluctuations in electrical signals and provide filtering in power supplies to remove unwanted noise.



Figure 7

3. Inductor:

Inductors are passive components that store energy in the form of a magnetic field when current flows through them.

- **Function:**
  Inductors oppose changes in current, making them useful for filtering high-frequency signals or storing energy in devices like transformers or motors. They can smooth out the ripple in power supply circuits and are often used in combination with capacitors to form tuned circuits.



Figure 8

4. Diode:

A diode is a semiconductor device that allows current to flow in one direction only, blocking current in the opposite direction

- **Function:**
  Diodes are primarily used for rectification, converting alternating current (AC) to direct current (DC). They are also used to protect circuits by preventing reverse current, which can damage sensitive components.

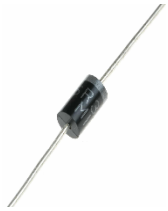


Figure 9

5. Relay:

A relay is an electrically operated switch that uses a small current to control a larger current in another circuit. It allows one circuit to control another without a direct electrical connection between the two.



Figure 10

- **Function:**

  Relays are commonly used for electrical isolation between high and low-voltage circuits. They also protect by allowing circuits to handle high-power loads that a simple switch might not handle safely.

-

6. Switch:

A switch is a simple device used to control the on/off state of an electrical circuit.

- **Function:**
  Switches allow or stop the flow of current in a circuit. They can be manually or automatically operated and are used in virtually every electrical device for controlling power flow.



Figure 11

7. Transformer:

A transformer is a device that transfers electrical energy between two or more circuits through electromagnetic induction. It is primarily used to step up

(increase) or step down (decrease) voltage levels.

- **Function:**
  Transformers play a critical role in power distribution by ensuring that electricity is transmitted efficiently over long



Figure 12

distances. They are also used in electrical devices to isolate different parts of a circuit and ensure safety.

8. Fuse:

A fuse is a safety device that protects an electrical circuit from overcurrent by melting and breaking the circuit when the current exceeds a safe level.

- **Function:**
  Fuses are essential for preventing damage to electrical components and reducing the risk of fire due to overheating. When a fuse blows, it interrupts the current flow until it is replaced.



Figure 13

9. Bridge Rectifier:

A bridge rectifier is an arrangement of four diodes connected in a bridge configuration. It converts alternating current (AC) into direct current (DC).

- **Function:**
  A bridge rectifier allows current to flow in only one direction, effectively transforming AC into DC. It is a key component in power supply circuits used in chargers and adapters.



Figure 14

○ Electronic devices:

An electronic device is a gadget employing semiconductive elements, such as silicon, Germanium, and the like, to facilitate electrical flow and execute specific functions. Take a mobile phone, for instance; its circuitry is predominantly composed of semiconductor materials. Upon activation, the mobile phone adeptly transforms electrical energy into intelligent operations, encompassing touch interactions, gesture control, networking capabilities, and memory management.

Electronic devices, on the other hand, exclusively function with Direct Current (DC). The voltage variety for digital gadgets is typically within the 'mV' to 'V' range, with common ratings like 5V, 12V, 24V, and many others.

Electronic gadgets, in assessment, consciousness on controlling voltage and present day. This management is accomplished through using lively components like transistors and diodes, taking into consideration the manipulation and amplification of electrical signals.

Electronic gadgets, exemplified by using modern technology like cellular phones, have the capacity to make short choices. This ability for decision-making units, digital devices aside, in terms of functionality.

So, it has common parts with electric devices, with some little differences between them, which are:

1. **Integrated Circuit (IC):**
   An IC is an electronic component that contains many connected circuits inside a single package, sometimes called a chip. It often has a 3- to 5-digit number printed on it (plus letters) that identifies its function or manufacturer.

- **Function:**



Figure 15

ICs perform a wide variety of tasks depending on their type: power regulation (e.g., 7805, LM317), logic operations (e.g. CD4026, CD4017), signal generation (e.g. NE555, LM741), switching (e.g., CD4066), or driving circuits (e.g. ULN2003). They provide compact, efficient, and reliable circuit implementations by integrating many components.

• **Package Types:**
– DIP (Dual In-Line Package), e.g. DIP8 has 8 pins.
– Metal-can (e.g., TO-99), often used for high-quality op-amps or precision ICs.

2. Transistor:

A transistor is a semiconductor component with three terminals that can control the flow of electrical current. It operates similarly to a switch, but instead of being controlled manually, it is controlled by a small electrical current.



Figure 16

- **Function:**

A very small current applied to the base (B) terminal can control a much larger current flowing between the collector (C) and emitter (E). Transistors are used in a wide range of electronic applications such as signal amplification, electronic switching, logic circuits, and switching-mode power supplies. They are commonly available in two main types: NPN, which works with positive voltage, and PNP, which works with negative voltage. Transistors come in different shapes and packages, and each one is identified by a number printed on its body.

3. Sensor:

A sensor is a device that detects and responds to input from the physical environment. Its main components include a receptor, transducer, signal conditioning electronics, and an output indicator. The receptor collects environmental input such as light, heat, sound, or magnetic fields. The transducer converts this input into an easily measurable form, usually an electrical signal. The signal conditioning circuitry amplifies or modifies the signal to match application requirements. Finally, the output indicator presents the result in a human-readable form, such as lights, sounds, or display readings.

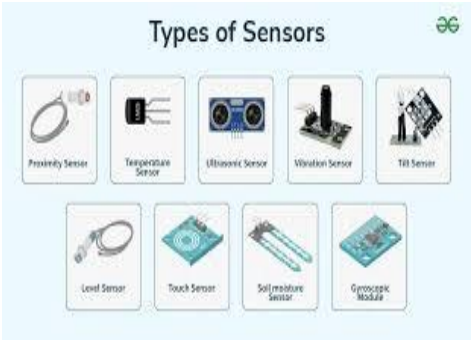


Figure 17

- **Function:**
  Sensors operate on the principle of taking an input and producing a related output. Steps include:

- Receptor: Detects environmental phenomena like temperature, light, motion, or magnetic fields.
- Transduction: Converts the input into another form of energy, e.g., thermal energy to electrical energy.
- Signal Conditioning: Amplifies and processes the signal, including filtering and digital conversion.
- Output: Provides a readable signal such as voltage, current, frequency, or digital code.
- Feedback (optional): Some sensors include feedback loops to improve accuracy or compensate for environmental changes.

• Types / Sensing Principles:

1. Mechanical: Detects deflection, stress, strain, or frequency/resistance changes to sense mechanical quantities.
2. Thermal: Uses temperature-dependent properties of materials (e.g., thermistors, thermocouples).
3. Electrical: Detects voltage, current, charge, or conductivity changes.
4. Magnetic (Hall Effect Sensors): Uses the Hall Effect to measure magnetic field strength, direction, or position. Hall Effect sensors are commonly used in:

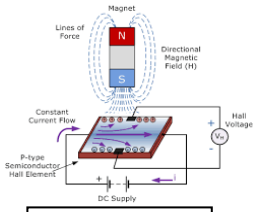


Figure 18

- Linear Hall Sensors: Provide a continuous output proportional to magnetic field strength.
- Digital (Switching) Hall Sensors: Output a binary signal when a magnetic field passes a threshold.
- Latching Hall Sensors: Maintain their output state until a magnetic field of opposite polarity is applied.

- Integrated Hall Sensor ICs: Combine signal conditioning and output stages in a single chip.

5. Radiant: Detects light, infrared, or radiation based on photoconductivity, photoemission, or bolometric effects.
6. Chemical: Uses electrochemical reactions, selective membranes, or catalytic reactions to detect specific chemicals (e.g., ion-selective electrodes).

   All the above are main components for all devices, so our idea depends on this stage of research and detection.

2. Microcontroller:

A microcontroller is an integrated circuit that is housed within each component that it needs to perform the necessary operations and that can perform a particular task routinely without requiring another boom. It contains a microprocessor, memory units and input-output interfaces, analog-to-digital conversion (ADC), pulse width modulation (PWM) and various control and communication modules.

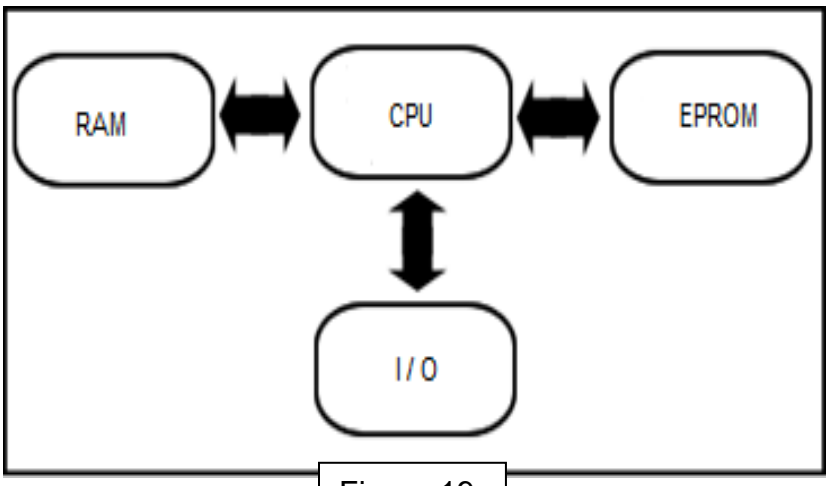


Figure 19

Besides the arithmetic-logical unit (ALU), special registers, processor control unit and processor core in the processor, the microcontroller has memory units with special functions. They can be divided into fixed and temporary memories. Random access memory (RAM) stores the information that the processor needs during its operation, while ROM, PROM, EPROM, and EEPROM memories, which do not change the contents, store information such as command set, program and program data as long as they are not reprogrammed. Input and output are special recording areas that allow the microcontroller to receive data from external units and send data .

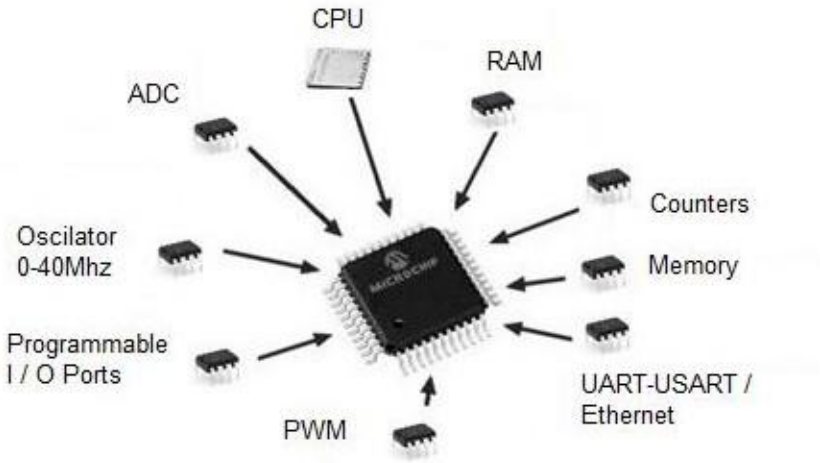


Figure 20

- **Microcontroller Types:**

Microcontrollers are generally classified according to processor characteristics such as processor architecture, processor word processing length, processor clock operating frequency, and so on. While RISC architectural microprocessors are generally preferred in electronic applications such as control and automation, CISC architectures are used in microcomputer applications such as signal processing and embedded systems. Some important features that distinguish microcontrollers from each other are listed below.

•Processor Architecture: type of application and command set used
•Processor word length: length and type of data to be processed
•Processor clock frequency: The processing speed of the code to be executed
•RAM and ROM capacities: The area covered by the program and program data

•Input / Output units: Data exchange and communication with external media
•Built-in enhancements: Special enhancements required

- **Operating conditions: Supply voltage, Microcontroller Selection:**

It may be confusing for beginners to decide which microcontroller to use when developing an application. While choosing the right microcontroller helps the user in the later stages of implementation, it can cause the application to be blocked at some point and returned to the planning stage with the wrong microcontroller. The following steps can be followed to select the microcontroller.

- **List the equipment required for the application:**

Specify the hardware needed for communication interfaces (USB, SPI, UART etc.) and special processes (ADC, PWM, DSP, FPU). Also list the port connections you will use for the input / output elements such as sensor, LCD display, driver and relay.

- **Observe your software architecture:**

Specify the length of the program to be operated and the operating speed. The content of the codes to be operated and the frequency of operation will determine your hardware requirements.

- **Specify the hardware architecture:**

Determine your hardware architecture based on the length of the data to be operated and the variety of instructions.

- **Specify the memory requirement:**

Specify the area of the program and program data that is required for the application. Calculate the RAM required to operate this program.

- **Cost and energy consumption analysis:**

Check the cost and availability of the equipment to be used. Check your application's energy consumption, operating frequency and other variables.

- **Examination of software development environment:**

Check the development environment and programming languages provided for the hardware you have selected.

- **Exploring the compiler and other development tools:**

Search the programming interfaces and tools to be used.

- **System testing and application experiments:**

If possible, conduct small practice tests to check that each system you use will be compatible with your system and work

When choosing a microcontroller development card, it is necessary to consider several features such as card interface, intrinsic hardware, program development interface, operating voltage, input/output numbers, as well as processor power and capacity. In addition, care must be taken to ensure both hardware and software compatibility of the extra equipment supporting the development board. When choosing a development card, the following must be observed:

- Supported processor types (PIC, AVR, AMR etc.).
- Communication interfaces (RS232, USB, UART, SPI, WiFi, Bluetooth etc.).
- Additional equipment on the card (LCD, Keypad, Button, Motor driver etc.).
- Electrical properties of the card (such as supply voltage, maximum current, and electromagnetic compatibility)

- Software flexibility and supported compilers (programming languages and compiler variants).

- Expandable properties of the card according to the application (additional module and connection interfaces).

- Physical properties of the card (size, weight, working temperature etc.).

- Attention should be paid to features such as price, availability and after-sales software support of the card.

3. Velocimeter device:

- What is the velocimeter?

- It poses a crucial role as it measures and displays the speed of vehicles or moving bodies.
- It has an educational purpose as it simulates many concepts in mechanics and physics as velocity, displacement and may give acceleration by some adjustments, so small module of it may be in school labs.

- How does it work?

The answer depends on what type of speedometer is used, as it has two main types:

⇒ Analog speedometer:

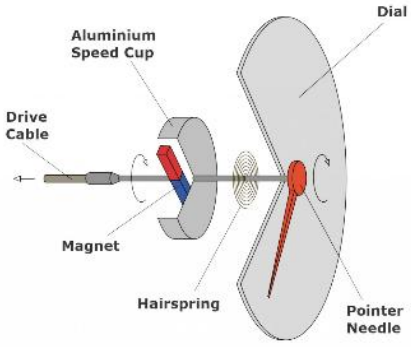


Figure 21

- It is the old version of speedometers, as it depends on a mechanical idea that was implemented in the last century, and it simply operates through intricate mechanical connections. A needle, connected to the outer shaft or gearbox, displays speed on a dial. The process begins with the driveshaft spinning the wheel and the connected speedometer cable. As the cable moves, it forces a magnet inside the speed cup to rotate, generating torque. This torque

propels the speed cup to spin at the same speed. However, a hairspring limits the rotation, allowing the pointer to move across the dial, indicating the speed accurately.

- It is also coupled with an odometer, which calculates the distance travelled

- **Advantages:**
  - Robust and less prone to malfunction
  - Cost effective option
- **Disadvantages:**
  - Less precision and accuracy
  - limiting personalization according to driver preferences.

⇒ **Digital speedometer:**

- It is the most recent way to have accurate results while measuring the velocity of the vehicles and it mainly depends on using sensors and the electronics technology of daily life as it operates seamlessly through sensors and chips. Magnets affixed to the outer shaft interact with magnetic sensors on the opposite side. When the magnets pass the sensors, they generate electric pulses in the circuit. These pulses are translated into speed and displayed on an LED screen. Since the digital system counts wheel rotations, it can also present precise odometer readings, making it a more accurate and convenient alternative.

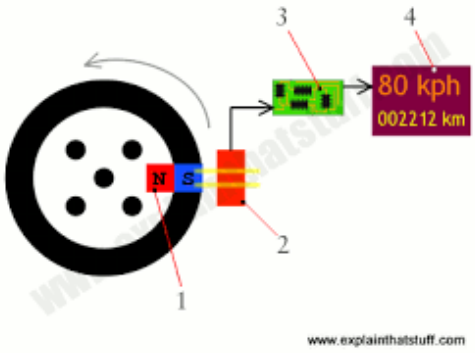


Figure 22

- **Advantages:**
  - Digital speedometers provide highly accurate and precise speed readings.
  - Digital speedometers can be customized and configured to multiunit and functions.
  - Can be integrated smoothly with cars' electric systems.

- **Disadvantages:**
  - Dependency on power and electric systems of cars
  - Tend to be more expensive because of its features.

4. The motion of bodies:

There are variable types of motion of bodies, and they depend on variable conditions and measurements like velocity, distance and acceleration which classify motion into:

○ Constant velocity:

In this type of motion, the body moves with uniform velocity and acceleration is zero as the body moves the same distance through the same intervals of time.

○ Non uniform velocity:

In this condition the body is moving with different rates as it move different displacements in different intervals of time so this motion can by classified into:

- **Constant acceleration:**

The body is moving with variable velocity may decreasing or increasing but the rate of change of velocity is same for the body, in this condition it involves the kinematics equations so calculate different quantities in terms of other quantities as they showed below:

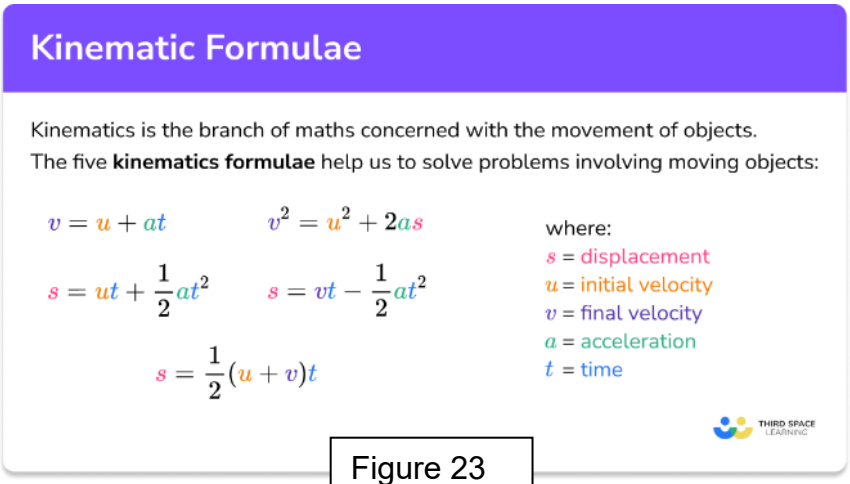


Figure 23

These formulas used to calculate these quantities which displacements, time, acceleration, initial and final velocities

- **Non uniform acceleration:**

- In non-uniform acceleration there is a variation in average acceleration of the body through different points along the path of the body , these differences may be in the magnitude or in the direction of the body movement , that means that the velocity of the body change in different rates along the path of it Thus, it is dependent on both velocity and time.

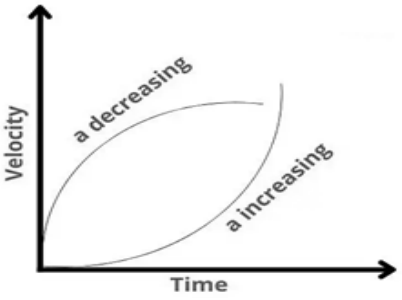


Figure 24

- Variable acceleration is directly related to time, as changes in acceleration occur over specific time intervals.
- When dealing with variable acceleration problems, it is important to consider how velocity and displacement are expressed as functions of time. By understanding these relationships, we can effectively solve examples and analyse the impact of changing acceleration on an object's motion.

  So, the graph simulation will be an effective way for getting the acceleration at different moments, as the slope of the velocity time graph represents the acceleration of the body, as shown below:

The graph illustrates the slope of decreasing or increasing non-uniform acceleration of the body, which is represented by a curved line, as they are not uniform.

IV. Prior Solutions

❖ What is e-waste?

- Waste electric and electronic equipment (WEEE) covers any electrical or electronic equipment that requires an electric current, a battery, or solar energy to operate and has reached its end of life and has been discarded. This includes both working and broken items. WEEE is particularly dangerous

due to toxic components that may leach to the environment when landfilled. In Egypt, WEEE is divided into eight categories:

1. IT and telecommunication equipment. Desktops, PCs, tablets, phones, printers, servers, copying equipment, monitors, telecommunication equipment, etc.
2. Screens and monitors
3. Temperature exchange equipment. Smoke detector, thermostat, measuring equipment, and other control instruments.
4. Light sources (lamps). Fluorescent lamps, sodium lamps, other lighting equipment, etc.
5. Medical devices. Radiotherapy, cardiology and dialysis equipment, other medical equipment (labelled).
6. Photovoltaic modules
7. Other large appliances
8. Other small appliances

- E-waste is a rising global issue:
  - The rapid pace of technological advancement has led to increased consumption of electronic devices, many of which are challenging and costly to recycle efficiently, so e-waste is a growing global issue that must be managed to keep the environment clean, as e-wastes include many hazardous materials and toxic metals that are found in its components.

  - electronic waste (e-waste). E-waste has become one of the largest and fastest-growing sources of pollution globally, driven by the booming electronics manufacturing industry. It was reported by the Global E-Waste Monitor (2024) that 62 million tons of e-waste were produced in 2022, and the generation of e-waste is on track to rise to 82 million tons in 2030. Waste electrical and electronic equipment (WEEE) poses a serious issue due to its complex mix of materials, making recycling both difficult and resource intensive.

❖ How to solve this problem?

- No single strategy will resolve this issue, but coordinated efforts that combine technological advancements, policy reforms, and global collaboration are crucial for effective long-term management.

- The need for advanced recycling technologies remains critical, as the lack of efficient and scalable e-waste recycling facilities contributes significantly to the growing global waste crisis. This issue has been widely documented, drawing attention from policymakers, engineers, and manufacturers alike. In recent years, mounting pressure to develop sustainable recycling solutions, particularly for composite materials, has intensified Composite materials, typically consisting of polymeric and metallic components, share recycling challenges like those of electronic waste. Various industries employ comparable recycling frameworks, highlighting the potential for cross-sector innovation. One such framework, the Technology Readiness Level (TRL), assesses the maturity of technologies and has been applied to evaluate the lifecycle of wind turbine blades to determine their recycling viability and cost-effectiveness.

- Implementing a similar framework for e-waste could significantly improve recycling strategies, emphasizing the importance of a well-structured, long-term plan for sustainable resource management.

• Effective recycling plan:

- An effective recycling plan is both a necessity and a luxury in addressing the current e-waste crisis. Given the diverse range of electronic waste, multiple recycling approaches are essential to handle various components effectively. Printed circuit boards (PCBs), among the most common components in electronics, are particularly challenging to recycle due to their complex mix of organic materials, glass fibres, and valuable metals.

- **The PCB recycling steps:**

  - The PCB recycling process generally involves three critical stages: pretreatment, physical recycling, and chemical recycling as shown in figure X. Pretreatment involves cleaning, disassembling reusable and hazardous parts, and mechanically separating materials through shredding and classification techniques. Physical recycling follows, focusing on material recovery through refining and purification processes. Chemical recycling is the final stage, often employing methods such as pyrolysis, thermolysis, and gasification to break down remaining components and extract valuable elements. The complexity of PCB recycling highlights the pressing need for efficient, scalable recycling technologies to mitigate the growing e-waste problem.



Figure 25

- **Pyrolysis:**

  - Pyrolysis is a prominent chemical process for managing plastic waste, capable of converting plastics into valuable fuels, chemicals, and even new plastics. While promising, the technology still requires further refinement to optimize efficiency and scalability. During pyrolysis, long-chain polymer molecules are thermally decomposed into smaller hydrocarbon fractions, producing oils and gases that serve as valuable precursors for fuel and chemical production. These outputs can be further processed to generate usable fuels and industrial gases. Although the process involves high operational costs, the value of the recovered oils and gases presents a significant opportunity for pyrolysis to become a more economically sustainable method of plastic recycling.

- **Composite and Fibrous:**

  - Composite and fibrous materials are among the most difficult to recycle due to their complex structures and manufacturing processes. Carbon fiber, a widely used fibrous composite material, is produced through a thermosetting process where heat hardens the fibers into a lightweight, rigid structure with aligned fibers. However, conventional recycling methods often degrade the material, leaving the fibers misaligned and fluffy, which significantly reduces their structural integrity and performance.

- **Thermolysis**:

  - Thermolysis, an advanced chemical recycling technique, offers a more effective approach by preserving the fiber's alignment and structural properties. By carefully adjusting temperature and pressure conditions, thermolysis can be applied to various types of fibrous composite materials, making it one of the most mature and efficient recycling methods for such challenging materials.

- **Gasification:**

  - Gasification involves the reaction of plastic waste with a gasifying agent, such as air or steam, at elevated temperatures ranging from 500 to 1200 degrees Celsius. This high-temperature process breaks down plastics into a flammable synthetic gas, commonly known as syngas. Syngas, primarily composed of hydrogen and carbon monoxide, can be used as a fuel for electricity generation or as a feedstock for producing chemicals and other materials. While it is not typically used as a direct fuel source like gasoline, its versatility makes it valuable for energy recovery and industrial applications. Gasification has been extensively developed and refined over the years, making it a highly effective method for reducing plastic waste.

- These methods offer effective strategies for managing existing plastic waste and guiding future decisions on plastic disposal. However, a significant challenge remains—addressing the plastic waste already polluting our environment. Biodegradation is a natural process in which microorganisms break down materials into carbon dioxide, water, and other simpler compounds. While all materials undergo biodegradation over time, the rate varies greatly.

• Biodegradation and Biodegradable Materials:

- Organic materials, like paper and cloth, decompose relatively quickly, while plastics and metals persist for much longer, contributing to long-term pollution. This disparity explains the increasing shift toward biodegradable alternatives, such as paper straws, cups, and containers, over traditional plastic products. Though biodegradation primarily involves biological activity breaking down materials, both biotic (biological) and abiotic (non-biological) factors influence the process. Factors such as temperature, humidity, and the presence of oxygen play critical roles, alongside microbial activity. By understanding and manipulating these factors, it is possible to accelerate the biodegradation of persistent materials, like plastics, presenting a promising pathway to mitigate plastic pollution more effectively.

- Various factors, including temperature, climate, mechanical forces, and chemical conditions, influence the biodegradation process. In natural biodegradation, the sun's heat and ultraviolet (UV) rays play a crucial role in breaking down materials to a state where microorganisms can metabolize them fully. By studying these natural processes, we gain valuable insights into what makes a material more biodegradable. This knowledge enables us to design new materials, such as bioplastics, that degrade more easily and efficiently. Bioplastics, derived from renewable sources, represent a promising solution for reducing e-waste. These materials maintain many of the essential properties of traditional plastics, such as durability and

versatility, while offering the added benefit of semi-rapid biodegradation once discarded. Bioplastics are already widely used in everyday applications, including food-grade materials, and their integration into electronic components could significantly mitigate e-waste challenges. The growing emphasis on bioplastics provides a hopeful pathway for more sustainable waste management practices in the future.

- **Biotechnology and Metal Recovery:**

  - Increasing investments in biotechnology are essential for minimizing the improper disposal of electronic waste. Governments and large corporations should prioritize these efforts to promote sustainable waste management practices. Emerging biotechnological approaches, such as bioleaching and biosorption, have shown significant promise in the recovery of valuable metals from electronic waste. Bioleaching, for example, has been investigated for its effectiveness in extracting copper from e-waste. A study on metallurgical processing using bioleaching reported a copper recovery efficiency of 24% based on the total copper content, highlighting the potential of the process but also the need for further optimization to maximize metal recovery. Complementary electrochemical methods have also demonstrated effectiveness in copper recovery. In a study using ammonia-based solutions, electrodeposition was explored to determine how copper concentration and ammonia salt levels influenced the process. The results indicated that higher copper concentrations led to greater ion availability for electrochemical charge transfer, slightly reducing the charge transfer coefficient but significantly increasing the transfer rate constant. This ultimately accelerated the overall metallurgical processing efficiency.

- **Deep Eutectic Solvents (DESs):**

  - Biodegradation can be accelerated using a combination of chemical and mechanical processes. One effective and relatively inexpensive method for chemically breaking down plastics is the use of deep eutectic solvents (DESs). DESs are mixtures commonly composed of ammonium, phosphonium, or sulfonium salts, combined with a hydrogen bond donor and a hydrogen bond acceptor. Depending on their composition, DESs can be categorized into various types based on their chemical properties and applications. DESs are particularly effective at dissolving plastics due to their strong solvent capabilities. Their unique chemistry, often rich in Lewis's acids, allows them to act as catalysts for a wide range of organic reactions. Lewis's acids, which accept electron pairs, enhance the degradation process by promoting the breakdown of polymer structures, making DES a promising tool for accelerating plastic biodegradation.

- **AI and IoT Integration:**

In addition to biotechnology, the integration of AI into IoT (Internet of Things) provides a hopeful move toward ecological balance, helping to address the growing problem of e-waste. IoT devices, which transmit data on the internet or other communication networks for certain applications, may be highly valuable in improving waste management. IoT-enabled smart tracking systems can improve waste collection efficiency by monitoring device lifespans and guiding users toward responsible disposal options and guiding people to responsible disposal choices. Additionally, IoT-enabled recycling facilities can streamline sorting and processing through complete automated material identification, considerably increasing the recovery rate of valuable resources. As technology rapidly advances, the application of AI and IoT to improve e-waste management will be crucial for dealing with the increasing demand for sustainable recycling solutions.

Chapter II

(Generating and Defending a Solution)

I. Design requirements

II. Selecting of solution

III. Selecting of prototype

I. Design Requirements

Design requirements define the functionality, usability, and constraints of our prototype. Clear requirements ensure it meets user needs and delivers a strong, effective solution.

1. Students will build a single scientific prototype instrument using at least one type of laboratory, whether it's Biology, Chemistry, Earth Science, Physics, or Fabrication Labs.

2. If the chosen scientific instrument possesses only one measurable function, the student is required to present outcomes from two actual laboratories' applications. Conversely, if the instrument has multiple measurable functions, at least one real laboratory application must be demonstrated.

3. If the team chooses to build a thermal instrument prototype, it must include an additional function compatible with the instrument's primary function.

4. The prototype should incorporate multiple components sourced from old, used, or discarded electrical devices. At least one must be an electrical component, while the remaining components will be chosen by the capstone team based on the specific functional application of the selected instrument. Additionally, several items may be chosen from a different component category.

5. If your prototype includes sensors, they must be calculated using standard methods or instruments, and the calibration process will be documented on the evaluation day.

6. The internal and external housing/cover of your prototype instrument should be made from a transparent material that may be extracted from the old devices, manufactured, or purchased.

7. The team is required to demonstrate the prototype's primary characteristics, which include the necessary voltage, the maximum drag electric current, and the maximum operating power. In addition, other selected characteristics/requirements should be presented based on the instrument's functions.

8. The team is required to present a schematic diagram that illustrates both the external and internal structural design of the prototype instrument. Additionally, it should include the electric circuit diagram if utilized in the design or necessary for the operation of the prototype instrument.

9. Students should present the results of the prototype instrument trial test and its application/s using graphical fitting methods with at least five readings.

10. The team should provide a video for the components' extraction, prototype building, trial tests, prototype characterization, and the instrument application/s.

11. The team should provide calculations for accuracy (not exceeding ± 5%) and precision, either compared to a functional device (or a known standard with relevant scientific references).

12. Functional prototypes must be testable, workable, reliable and portable on the exhibition day. The team should also bring a hard copy of each team member's logbook.

⇒ We must use this:

1. Only old, used, or discarded electrical devices may be used.

2. You must adhere to all safety rules and documents during each step of building a prototype instrument, especially during the component extraction

step and instrument testing. Be careful of the high-voltage equipment present in some old electrical devices.

3. All processes, from component extraction to final prototype application testing, should be conducted within the school laboratory.

4. The prototype must be testable, workable, and portable at any time and place. “If necessary, the team should bring spare parts for the easily damaged components on the evaluation day.

5. No equipment or materials that belong to the laboratories can be removed from the laboratories at any time. You must conduct those tests that require this equipment in the labs. Failure to do so will result in a reduced grade.

6. It is not allowed to use materials that contain radioactive elements or hazardous chemicals.

II. Selection of Solution

Our capstone idea talks about the recycling of electronic waste and salvaging it. There were many ways to recycle old electronic gadgets, each with its own advantages and disadvantages, and with each concept needing specific environmental requirements.

After looking at multiple points of view on the problem and the challenge of our capstone, and after searching and analysing many different prior solutions, we decided to pursue the current idea of our capstone, which is recycling waste electronics and electric devices, and reusing the extracted components to construct a system that measures the speed, acceleration, and the distance covered by a discarded RC car and display graph of the readings on a website

Advantages:

- **Enhancing the learning experience:**

    - As the system uses a non-serious approach to teach the concepts of speed and acceleration, using an RC car that can be played with, this would make the students a lot more eager to learn such concepts.

    - As the system displays the readings on a website using a graph, this makes it easier for the students to visualize the results and could potentially improve their graphing skills in addition to clarifying the concepts mentioned.

    - The system is intended to be implemented in the physics lab, solidifying the students' understanding of the physics concepts.

- **Versatility and low operation standards:**

    - The system does not need a power supply of 220V, just 6 AAA batteries, and a laptop can power it effectively.

    - The system is lightweight, portable, and easy to disassemble and reassemble.

    - To display the graph, just connect the ESP 32 CAM to the same WIFI the user's laptop is connected to.

- **Contribution to the recycling efforts:**

    - Since the system mainly relies on discarded, extracted, and recycled components, it directly contributes to the national recycling efforts. It

is also a reminder that nearly anything can be recycled, even long-gone electronics.

- Not only does the system use recycled electric components, but it also uses discarded wooden planks as a designated track for the RC car.

- **Low-budget friendly:**

  - As the system needs only recycled materials and components, it roughly costs about 1300LE to be fabricated, which is cheap compared to the other expensive lab apparatuses.

Disadvantages:

- **Unexpected Errors:**

  - Unexpected errors could occur, as the system utilizes an ESP32 CAM as a Wi-Fi module to send the results and the data to the user's laptop to display graphs, which could malfunction as it easily fills with cache.

- **Using cables:**

  - As the system needs a cable that connects the ESP32 CAM to the users' laptop to power it, as it is forbidden to power it using batteries, it could lead to irreversible damage to the ESP.

III. **Selection of Prototype**

Based on the problem identified above with regards to rising electronic waste and the need for affordable and efficient scientific equipment in laboratories, our team chose the construction of a recycled digital velocimeter as a suitable project option. The project aims to construct a device that can detect the distance, speed, and acceleration of a moving object using recycled digital components, among other functions.

The prototype chosen is a combination of an old remote control toy car, which serves as the moving object, paired with an ESP32-CAM microcontroller obtained from an old smart camera, and an ultrasonic sensor obtained from old electronic devices. This setup enables the real-time calculation of motion parameters within the stipulated requirement of utilizing recycled electronic devices. The reason why ESP32-CAM microcontrollers were used is due to their Wi-Fi connectivity, processing capabilities, and sensor data acquisition functionality, which make them ideal for functioning as the control devices in a system.

This prototype was selected because it meets a variety of design criteria at one time. First, this prototype gives a variety of measurable functions such as distance, speed, and acceleration, ensuring it can be used in a variety of physics lab experiments. Secondly, this prototype enables lab results to be recorded through wireless transmission to a web platform where results are presented both in table and graph formats, facilitating understanding of both motion and graph analysis concepts. Thirdly, this prototype runs at a low voltage, is light in weight, and is safe for lab experimentation.

Moreover, the outer casing of this system is transparent, making it easier to see all components inside, thus making it more educational. Also, using recycled materials not only makes this system inexpensive but also promotes sustainability. As far as the cost of commercial motion sensors and speed meters is concerned, this prototype system is far cheaper and can be customized. Hence, this prototype was chosen because it successfully incorporates elements of sustainability, accuracy, education, and feasibility into a working system.

Chapter III

(Constructing and Testing a Prototype)

I. Materials and Methods
II. Test Plan
III. Data Collection
IV. Materials and Methods

I. Materials and Methods

Materials:

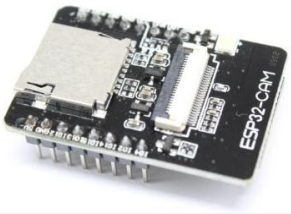
Item	Usage	Picture	Item	Usage	Picture
Esp32cam	From an old camera and it is the mind of the project		Old toy car	We put inside it all the components	
Ultrasonic sensor	We used to measure distance		Old track from wood	We used this track to put the car on it	
Jumber wires	To connect components		Micro usb cable	To transfer code to esp32cam	
Iron nails	For the wood track		Wax gun	To stick the components	

Table 1

Methods:

Various methods were executed to construct this prototype. Things like accurate measurements, thorough selection of materials, and safety precautions were put in perspective during construction.

Materials mentioned in table 1 were utilized in constructing this contraption. Every element was chosen after deep research on it. This helped us substantially during construction, as every element role and location was known.

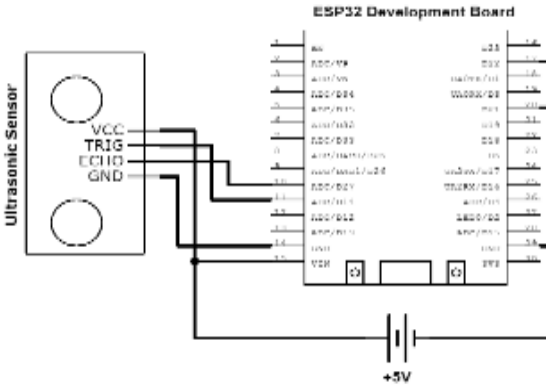


Figure 26

The construction of the prototype (speedometer) requires achieving design requirements and configuring the electric circuit first. We make a schematic diagram of the project to simulate the prototype and its electrical connections.

Steps of installation:

1. A deep search was conducted to find old electric devices that include a microcontroller.

2. After locating an old smart camera that includes a microcontroller called esp32 cam, the microcontroller was extracted from it to be the “mind” of our contraption.

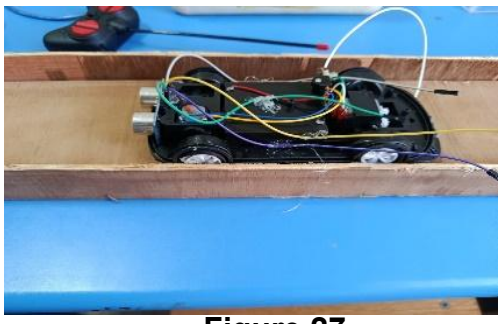


Figure 27

3. A calibrated ultrasonic sensor is used to detect the distance covered by the body.

4. Old jumper wires were used to connect these components.

5. The microcontroller was coded to take the readings from the sensor to return the results of the sensor detection.

6. Programmed a website with (HTML, java, CSS) to show the results in various ways, such as tables and graphs.

7. Stuck all these components on the body of an old toy car to be the detected moving body

8. Design a small track with old wood Remnants, its length is about 122 ± 2cm, it will be the road of the moving body (toy car) in our prototype.

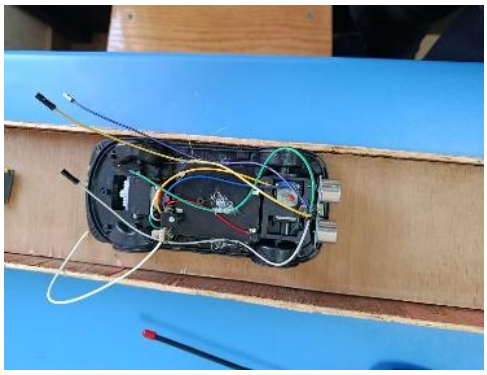


Figure 28

II. Test Plan:

Test Plan 1:

To start, before using the ultrasonic sensor in our system, the Hall Effect was used primarily to execute the same function. However, as shown in Table X, it proved to be unable to measure the distance covered by the RC car accurately. This test plan proved that the hall effect sensor was not suitable for our system, as it generated an

error percentage that is well over the accepted limit (±5%) mentioned in the design requirements.

Trial	Real Distance(m)	Distance Measured(m)	Error percentage
First Trial	0.98	1.35	37.8%
Second Trial	1.02	0.79	22.5%
Third Trial	1	1.24	24.0%
Fourth Trial	0.94	1.17	24.4%
Fifth Trial	0.97	1.42	46.3%

Table 2

Test Plan 2:

After the disastrous results from Test Plan 1, some changes were made, including the replacement of the hall effect sensor. These changes were:

- The installation and calibration of the ultrasonic sensor according to the location conditions.
- Attaching the whole system to the moving body (old toy car).
- Putting the car on the track after being polished.
- Checking the website before starting the test.
- Moving the car with the remote-control forward in a straight line on the track, not back and forth to avoid potential errors.

After these changes, the results, which are shown in Table 3, improved drastically.

Table 3

Distance	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average
Manual measuring value	1.02 m	1 m	1.01 m	0.98 m	0.99 m	1m
Measured value by sensor	1 m	0.98 m	0.99 m	0.96 m	0.97	0.98 m

This test plan exhibits the accuracy and precision of the ultrasonic sensor and how it achieved an error percentage of about 2%, which is well below the accepted limit mentioned in the design requirements.

III. Data collection:

Tools:

Various tools were used to either measure or in the construction of the system, like:

Oscillating Saw:

The tool exhibited in Figure X is a saw that works using the concepts of oscillatory motion, as it only cuts hard objects, avoiding cutting soft tissues, deviating potential dangers.



Figure 29

Measuring tape:

It was used to measure the length of suitable wooden scraps that can be cut, as well as to measure the actual distance that the RC car covered during the test plans.



Figure 30

Avometer:

It was used to determine the voltage and current intensity suitable for the components to avoid supplying them with more than they could endure.



Figure 31

Glue gun:

It was used to attach the ESP32CAM, the ultrasonic sensor, and the jumper wires to the body of the RC car. This was done to avoid the components from falling off during the course of the system operation.



Figure 32

Screwdriver:

It was used in the extraction of the components, loosening them from their electronic boards. It was also used to fasten screws into the wooden planks to construct the track,



Figure 33

Results:

Negative results:

After executing our tests according to the test plan, initially, some negative results came out, as there were some issues with the used Hall effect sensor, as it either

skipped a whole revolution of the magnet attached to the spinning wheel or it was not compatible with our microcontroller, which affected the results' accuracy as shown in table 4.

Trial	Real Distance(m)	Distance Measured(m)	Error percentage
First Trial	0.98	1.35	37.8%
Second Trial	1.02	0.79	22.5%
Third Trial	1	1.24	24.0%
Fourth Trial	0.94	1.17	24.4%
Fifth Trial	0.97	1.42	46.3%

Table 4

Positive results:

So, after further failed trials, using other sensor, which was the ultrasonic sensor, was the best option to generate accurate results. Fortunately, the ultrasonic sensor returned accurate results with utter precision, as the error percentage was **±2%**, as by measuring the length of the track manually **5 trials** we found the average is **1± 0.02 m** and the system returned data exhibited that the average distance travelled of 5 trials **was 0.98± 0.02 m** As shown in **Table 5.**

Distance	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average
Manual measuring value	1.02 m	1 m	1.01 m	0.98 m	0.99 m	1m
Measured value by sensor	1 m	0.98 m	0.99 m	0.96 m	0.97	0.98 m

Table 5

- Then the other quantities, such as speed, which calculated according to **5 trials** readings of time and distance and acceleration, are calculated from the conducted graphs through website. As shown in **Table 6.**

Quantity	Trial 1	Trial 2	Trail 3	Trial 4	Trial 5
Distance	1 m	0.98 m	0.99 m	0.96 m	0.97 m
Time	1.72 s	1.67 s	1.75 s	1.6 s	1.5 s
Avg. Speed	0.581 m/s	0.587 m/s	0.566 m/s	0.600 m/s	0.647 m/s
Avg . Acceleration	1.5	1.48	1.56	1.6	1.58

Table 6

Chapter IV

(Evaluation, Reflection, Recommendations)

I. Analysis and Discussion
II. Recommendations
III. Learning Outcomes
IV. Conclusion
V. Citation

I. Analysis and Discussion:

On spotting the light on Egypt's grand challenges, especially recycling and reused electrical devices, these two are key drivers of environmental problems, pollution, public health risks, and a variety of other challenges that confront Egypt. In the process of recycling, our speedometer is effective as it helps us to reuse old components to construct a new device and allows for better scientific and technological improvement. Speedometers support physics labs such as measuring speed, acceleration and distance and time. Our speedometer reduces electric waste and enhances the efficiency of the technological environment; it may be possible to reduce climate change. Most regions and countries on the earth have problems with climate change, recycling, and pollution that come from electrical waste. Because of this, most people have been affected by these. Every prototype has problems, but our prototype has demonstrated a combination of exploiting strong points and avoiding weak points. We achieved results that go beyond most of our design requirements. Our prototype has strength points like low cost and its parts are easy to repair and enhance the educational system operations, as it has contributed to a deep understanding of some physics concepts such as speed and acceleration, also we made a website for our project, and it has all the results to give all information.

- **Old remote-controlled car:**

  An old toy RC car was dismantled to be the test subject of the project, of which acceleration, distance covered, and speed will be tested.

  The ESP32 CAM microcontroller includes Wi-Fi and Bluetooth functionality, Bluetooth Low Energy (BLE) communication, independent timers, analog-to-digital and digital-to-analog converters (ADCs and DACs), as shown in Figure 34. The ESP32 microcontroller includes two 240MHz cores, each with a Tensilica Xtensa 32-bit LX6 microprocessor.



Figure 34

Ultrasonic sensor:

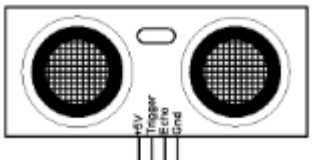


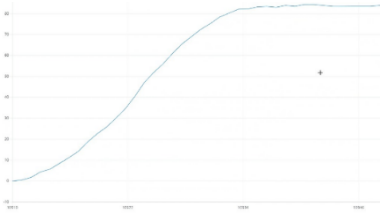
Figure 35

The ultrasonic sensor is a non-contact sensor that includes 4 main pins, VCC (5V), Trigger, Echo, and GND (Ground), respectively, as shown in Figure 35, measure distance with utter accuracy. However, the precision of its measurements is often low, as it is affected by external factors like humidity and air density. The TOF (time of flight) is the time taken by an ultrasonic wave to hit an obstacle and return. Ultrasonic-distance measurement with the TOF method is based on the following physical parameters: speed of sound in the air, environmental factors, and time of detection the reflecting wave, as illustrated in Equation 1.

$$D = \frac{V_S \cdot T}{2}$$

Equation 1: U.S. Distance Calculation

As mentioned above, there is a typical issue with using the ultrasonic sensor, which displays a stable graph without sudden dips and peaks, as exhibited in graph 1, which is meant to represent a static object, which reduces the overall precision.



Graph 2: U.S. distance readings

Equation 2 explains the change in sound speed due to temperature. C is the temperature in Celsius. Humidity impact is usually negligible in practice, but temperature must be taken into consideration to measure the distance changes up to the millimeter level. If the temperature changes by about 5 degrees Celsius, the distance changes by about 3 mm in 1000 microseconds (the speed of sound is 342.3 m/s at 20 degrees Celsius, and the sound distance traveled is about 343.2 mm in 1000 microseconds).

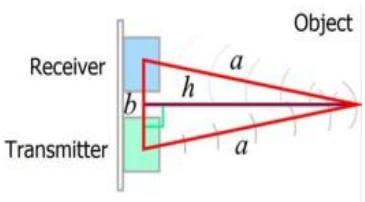


Figure 36: U.S working principle

A second test plan was conducted; however, the given result of 0.83m travelled distance was 0.81 m, as shown in graph 2, which is a 2% error percentage. This happened because the external factors, like humidity, temperature, and wind speed, were taken into

$$C_f = 2\pi r,$$

Equation 3: The circumference of a circle

consideration as windows were closed and fans were turned off to reduce the noise that could disturb the accuracy.

The system, also After conducting the first test plan with the use of the hall effect sensor, which proved its ineffectiveness, as a measuring tape was laid on the table to measure the distance covered by the toy car, which was one meter, the sensor returned to a value of 0.7m. The reasons behind this 30% are that the Hall effect sensor was incompatible with the project overall, as it needed to have a magnet attached to the spinning wheel of the RC car to calculate the number of turns by counting the magnet pulses and, with it, determine the distance covered, however, a small magnet was bigger that it should be as shown in Equation 3 which is used to calculate the distance covered per revolution, which is the circumference of the wheel, the magnet's size made the sensor falter, as it sometimes skipped a pulse or did not complete a whole revolution. Two smaller magnets were attached in a desperate attempt to solve this issue, but with no success.

$$v = \frac{d}{t}$$

Equation 4: Velocity Formula

$$a = \frac{\Delta v}{\Delta t}$$

Equation 5: Acceleration Formula

Changing the sensor was the only option to reduce the error percentage. The ultrasonic sensor was the best option as it has accuracy by sensing the reflection of sonic waves that are cast from the transmitter as they bounce off an object and return to the receiver using basic geometry, as shown in Figure 36, and Equation 3, measured the velocity and acceleration 2% error percentage as shown in table 6, as both of them are dependent on the measured distance and time as shown in equation 4 and 5, and shows the accuracy of the ultrasonic sensor.

II. Recommendation:

There is no perfect prototype, and our prototype is no exception. We recommend some features and attributes that can have a major effect on the project, which could

help in improving the application and efficiency of our project. To turn this project into a final product, several suggestions should be considered as follows:

- Improve Measurement Accuracy: using better, more accurate sensors along with the processing unit is critical for a larger-scale building, considering its scarcity regarding small scale prototypes.

- Reinforce Mechanical Components: Strengthen the car chassis and overall car body and the track, ensure stable performance, and reduce errors considering high-speed testing

- Adopt adaptive design: use flexible sensor and hardware modules for replacements and customizations for educational lab experiments.

- Increase Sustainability Impact: adopt a well-studied technique of recycling waste materials (using a well-studied plan of which materials will be recycled and how to ensure having the best impact on earth)

- Ensure Reliability and Safety: to make a final product out of our prototype, it must meet several safety requirements and standards to withstand any crisis and save students from any fatal error.

III. Learning Outcomes:

A lot of benefits were gained from working on this project, such as helping us become better STEM students and improve our collaboration skills. This project increases innovation, critical thinking, and improves problem-solving skills. Here are the learning outcomes that aided us the most in creating, constructing, and testing our system.

Subject	Learning Transfer
PH.2.03	**Clarified the laws of electrodynamics and helped us determine the suitable voltage for each component using Ohm's law (V=IR).**
PH.2.04	**Introduced Kirchhoff's Law, which aided us in determining the voltage and current intensity of parallel-connected batteries.**
PH.2.06	**Facilitated the usage of the Hall effect sensor, as it discussed magnetic fields and helped in determining the suitable distance between the sensor and the magnet.**
MA.2.01	**Facilitated the graphing of the equations used and ensured the correctness of the results.**
CS.2.01	**Aided in determining the topological shape and connecting the website to the contraption via local cloud.**
CS.2.02	**Introduced theconcept of of databases and how to connect different tables toghter, which help in the construction of the site responsible for displaying the results.**
CS.2.03	**Facilitated the code writing, as it introduced the concepts of loops, variables, as branches.**
BI.2.01	**It introduced the Idea of selecting favourable features, as it discussed the concept of GMOs.**
ES.2.01	**Explained the concept of water re entering the its cycle, and shaped our approach of WEEE recycling.**
English	**Help in conducting this research as a whole, in writing the poster and portfolio academically.**

Table 7

IV. Conclusion:

This research tackles not only the rising concern regarding the disposal of electronic waste but also the incompleteness of accessible and reliable solutions in terms of modelling motion ideas in a physics lab in schools. The use of recycled parts from electronic trash, coupled with the ESP32 CAM microcontroller, proves that engineering through sustainability can indeed create educational technologies that are both effective and affordable.

The analysis of the ultrasonic sensing function regarding the time-of-travelled distance calculation provides the essential theoretical background in forecasting accuracy with remarkable consistency in terms of speed and acceleration. The analytical results are well supported by the experimental testing. The ultrasonic sensor yielded consistent and comparable results in the laboratory setting, and the calculated values of distance, speed, and acceleration closely matched the theoretically expected values of motion in the track. This confirms that this system provides effective real-time data that can be utilized in physics lab works. This current prototype compares favourably with the former solution that relied upon commercial motion sensors that are too expensive and unrepairable when compared with this environmentally responsible and affordable system that functions correctly and remains straightforward in terms of maintenance.

The results obtained confirm the effectiveness of the use of recycled electronic components in satisfying the requirements necessary for powering a complete and accurate motion-measurement system, ensuring physics lab access enhancement and reduction of e-waste. Results on the confirmation from both analytical and experimental results, the next level involves utilizing this information in designing recommendations.

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